



COMPETENCY STANDARD
FOR
QUALITY CONTROL & ASSURANCE

(LEATHER GOODS & FOOTWEAR SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for **Quality Control & Assurance** is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as a framework for providing industry-aligned training, ensuring that individuals meeting the prescribed standards through assessments are qualified to secure relevant employment opportunities.

This document has been specifically developed for training purposes under the Skills for Industry Competitiveness and Innovation Program (SICIP), aiming to address the skills requirements of the leather goods and footwear sector. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:**Generic Competencies (30 Hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LET-QCA-01-G	Carry out workplace interaction	1. Interpret workplace communication and etiquette 2. Read and understand workplace documents 3. Participate in workplace meetings and discussions 4. Practice professional ethics at work	15
SICIP-LET-QCA-02-G	Operate in a team environment	1. Identify team goals and work processes 2. Identify own role and responsibilities within team 3. Communicate and co-operate with team members 4. Practice problem solving within the team	15
Total Hour			30 Hrs.

Sector Specific Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LET-QCA-01-S	Carry out measurements and calculations	1. Plan and prepare 2. Obtain measurements 3. Perform calculations	10
SICIP-LET-QCA-02-S	Apply green practices in LET sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hour			20 Hrs.

Occupation Specific Competencies (310 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LET-QCA-01-O	Understand Quality Control and Assurance	1. Interpret quality 2. Interpret quality control and quality assurance 3. Identify QC and QA tools 4. Identify roles and responsibilities of quality personnel	20
SICIP-LET-QCA-02-O	Identify Materials, Machines, and Accessories for Leather and Footwear Production	1. Interpret leather, leather goods and footwear 2. Identify raw materials and chemicals 3. Identify cutting machine 4. Identify sewing machine 5. Identify lasting machine 6. Identify types of trim and accessories	25
SICIP-LET-QCA-03-O	Interpret Leather, Leather Goods and Footwear Making Process	1. Identify leather making process 2. Identify leather goods making process 3. Identify footwear making process	40
SICIP-LET-QCA-04-O	Identify Leather, Leather Goods and Footwear Faults and Defects	1. Identify types of faults and defects 2. Identify leather faults and defects 3. Identify leather goods faults and defects 4. Identify footwear faults and defects 5. Identify trim and packing faults and defects	30
SICIP-LET-QCA-05-O	Carryout Section Wise Quality Control	1. Perform quality control in sample 2. Perform quality control in cutting 3. Perform quality control in embellishment 4. Perform quality control in sewing 5. Perform quality control in lasting and finishing	100
SICIP-LET-QCA-06-O	Apply Root Cause Analysis	1. Analyze data and identify root causes	40

Code	Unit of Competency	Elements of Competency	Duration (hours)
	Techniques	2. Implement root cause analysis techniques 3. Implement corrective and preventive actions (CAP/PAP) 4. Prepare report analysis results	
SICIP-LET-QCA-07-O	Perform Final Inspection and Audit	1. Prepare for final inspection 2. Conduct final product inspection 3. Interpret technical audit system	55
Total Hours			0 Hrs.

COMPETENCY STANDARD: QUALITY CONTROL & ASSURANCE

Generic Competencies

UNIT OF COMPETENCY: CARRY OUT WORKPLACE INTERACTION	Nominal Duration: 15 hrs.	Unit Code: SICIP-LET-QCA-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out workplace interaction. It specifically includes the task of interpreting workplace communication and etiquette, Reading and understanding workplace documents, participating in workplace meetings and discussions and practicing professional ethics at work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a <u>courteous manner</u> to gather and convey information. 1.4 <u>Workplace procedures and matters</u> are comprehended.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted correctly. 2.2 Visual information/symbols/signage are understood correctly and followed. 2.3 Specific and relevant information are accessed from <u>appropriate sources</u> . 2.4 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time. 3.2 Meeting procedures and etiquette are followed. 3.3 Active participation is ensured, opinions are expressed and heard. 3.4 Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	4.1 Responsibilities as a team member are performed. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is maintained. 4.4 Inappropriate and conflicting situations are avoided.

Range of variables:

Variables	Range (may include but not limited to)
1. Courteous manner	1.1 Effective questioning 1.2 Active listening 1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette
2. Workplace procedures and matters	1.6 Notes 1.7 Arranging a meeting 1.8 Agenda 1.9 Simple reports such as progress and incident reports 1.10 Job sheets 1.11 Operational manuals 1.12 Brochures and promotional material 1.13 Visual and graphic materials 1.14 Standards 1.15 OHS information 1.16 Signs
3. Appropriate sources	1.17 Human Resources (HR) Department 1.18 Managers 1.19 Supervisors 1.20 Management Information System (MIS)

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace codes of conduct 1.2 Workplace communication and etiquette 1.3 Workplace documents, signs and symbols 1.4 Meeting procedure and etiquette 1.5 Professional ethics 1.6 Responsibilities of team member
2. Underpinning skill	1.7 Demonstrating workplace communication and etiquette 1.8 Interpreting workplace instructions and symbols 1.9 Performing responsibilities of a team member 1.10 Demonstrating active participation in workplace meeting 1.11 Applying professional ethics at work
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided:

	1.1 Workplace (simulated or actual) 1.2 Projector 1.3 Learning manual
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings 1.4 Performed responsibilities of a team member
2. Methods of assessment	Competency should be assessed by: 1.5 Written test 1.6 Practical Demonstration 1.7 Oral Questioning 1.8 Portfolio (Optional)
3. Context of assessment	1.9 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 1.10 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE IN A TEAM ENVIRONMENT	Nominal Duration: 15 hrs.	Unit Code: SICIP-LET-QCA-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the task of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and cooperating with team members and practicing problem solving within the team.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1. Roles and objectives of the team are identified and interpreted. 1.2. Roles and responsibilities of team members are identified and interpreted.
2. Identify own role and responsibilities within team	2.1 Personal role and responsibilities are identified within the team environment. 2.2 Reporting relationships are interpreted within team and external to team.
3. Communicate and cooperate with team members	3.1 Other teammates' tasks are identified and support provided when requested. 3.2 The team is encouraged through <u>sharing information</u> or expertise, working together to solve problems, and putting team success first. 3.3 Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1 Problems faced at the individual and team level are identified and showed insight into the root causes of the problems. 4.2 A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3 The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4 It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	Range (may include but not limited to)
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1. Sharing information	1.1. Agenda 1.2. Minutes 1.3. progress and incident reports 1.4. Operational manuals 1.5. Visual and graphic materials 1.6. Emails and SMS 1.7. Phone directory 1.8. Policy, procedure and standards 1.9. OHS information
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Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of team members 1.3. Personal role and responsibilities 1.4. Sharing information 1.5. Views and opinions 1.6. Problem solving within the team
2. Underpinning skill	1.1. Identifying team goals and work processes 1.2. Identifying own role and responsibilities within team 1.3. Communicating and cooperating with team members 1.4. Practicing problem solving within the team
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes 1.2 identified own role and responsibilities within team 1.3 communicated and cooperated with team members 1.4 practiced problem solving within the team
2. Methods of	Competency should be assessed by:

assessment	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: CARRY OUT MEASUREMENTS AND CALCULATIONS	Nominal Duration: 10 hrs.	Unit Code: SICIP-LET-QCA-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for carry out measurements and calculations. It specifically includes the task of planning and preparing, obtaining measurements and performing calculations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Plan and prepare	1.1. Work instructions are confirmed and applied to the job in hand. 1.2. Materials to be measured are identified as per job specification. 1.3. Appropriate <u>measuring device</u> is identified and selected based on materials to be measured. 1.4. Specifications are obtained and verified from relevant <u>documents</u> .
2. Obtain measurements	2.1 Method of obtaining measurement is selected and applied. 2.2 <u>Measurements</u> are obtained using appropriate device in accordance with job requirement. 2.3 Measurements, including area, volume, tolerance and clearance limits, are confirmed and applied
3. Perform calculations	3.1 <u>Calculations</u> , using basic operations, for determining material requirement are taken. 3.2 Appropriate <u>formulas</u> for calculating quantities are selected. 3.3 Quantities are estimated from the calculation taken. 3.4 Material quantities are calculated, confirmed and recorded within tolerances.

Range of Variables

Variable	Range
	May include but not limited to:
1. Measuring device	1.1 Measuring tape 1.2 Steel rule 1.3 Calculator 1.4 Sets square
2. Documents	2.1. Technical manuals

	2.2. Specifications 2.3. Sketches 2.4. Drawings 2.5. Charts 2.6. Photographs
3. Measurements	3.1. Length 3.2. Width 3.3. Weight 3.4. Tolerance
4. Calculations	4.1 Addition 4.2 Subtraction 4.3 Multiplication 4.4 Division 4.5 Area 4.6 Volume 4.7 Circumference 4.8 Cubic meter 4.9 Volumetric weight
5. Formulas	5.1 Fractions 5.2 Percentages 5.3 Mixed numbers 5.4 Conversions 5.5 Scales

Curricular Content Guide

1. Underpinning Knowledge	1.1 Measuring devices 1.2 Measurements 1.3 Simple calculation techniques 1.4 Appropriate formulas 1.5 Estimating quantities
2. Underpinning Skills	2.1 Identifying appropriate measuring devices 2.2 Carrying out measurements for appropriate device 2.3 Performing calculations using appropriate formulas
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Communication with peers, sub-ordinates and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Map/globe 4.3 Projector 4.4 Stationery 4.5 Learning manual.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 planed and prepared 1.2 obtained measurements 1.3 performed calculations
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES IN LET SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-LET-QCA-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in LET sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices in LET (Ready-Made Garment)</u> manufacturing are explained. 1.2 Key terms and symbols in LET production processes are identified. 1.3 <u>Sources of environmental impacts</u> during LET manufacturing activities are identified. 1.4 LET manufacturing activities use are explained. 1.5 <u>Ways of mitigating environmental impacts</u> in LET manufacturing are explained.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw material consumption in LET manufacturing are documented. 2.2 <u>Recyclable and non-recyclable items</u> in LET production processes are identified. 2.3 Procedures to reduce resource consumption in LET manufacturing are implemented. 2.4 Sustainable alternatives in LET manufacturing are explored and applied.
3. Implement waste management practices	3.1 <u>Different types of waste in LET manufacturing</u> are identified. 3.2 Hazardous waste in LET production is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Principles of green practices in LET (Ready-Made Garment)	1.1 Reducing energy consumption in LET manufacturing processes. 1.2 6R for waste management in LET production: 1.2.1 Refuse unnecessary materials and processes.

	1.2.2 Reduce resource consumption and waste generation. 1.2.3 Reuse materials within the production process. 1.2.4 Recycle materials for future use in LET manufacturing. 1.2.5 Recover usable resources from waste. 1.2.6 Repair damaged products or materials instead of discarding them. 1.3 Use of sustainable materials in LET production with low environmental impact. 1.4 Recycling of materials used in the LET manufacturing process. 1.5 Sustainable transportation methods for moving goods and materials within the LET sector.
2. Sources of environmental impacts	2.1 Air Pollution in LET Manufacturing 2.2 Water Pollution in LET Manufacturing 2.3 Soil Degradation in LET Manufacturing 2.4 Noise Pollution in LET Manufacturing 2.5 Waste Generation in LET Manufacturing 2.6 Resource Depletion in LET Manufacturing
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment in LET production processes. 3.2 Adopting renewable energy sources for LET manufacturing operations. 3.3 Implementing site protection measures to prevent pollution and reduce environmental. 3.4 Using reusable materials in production processes. 3.5 Choosing sustainable materials with low environmental impact for garment production. 3.6 Using noise-reducing equipment and scheduling production work. 3.7 Optimizing logistics and delivery to reduce transportation emissions and improve overall supply chain efficiency.
4. Recyclable and non-recyclable items	4.1 Recyclable Items in LET Manufacturing 4.1.1 Metal components. 4.1.2 Wooden materials. 4.1.3 Fabric scraps and textile fibers. 4.1.4 Plastic items, including buttons, tags, and packaging materials. 4.2 Non-Recyclable Items in LET Manufacturing 4.2.1 Paints, coatings, and dyes. 4.2.2 Contaminated materials. 4.2.3 Treated wood and other materials.
5. Different types of waste in LET manufacturing	5.1 Fabric waste from cutting and trimming processes. 5.2 Wood waste from packaging or display materials. 5.3 Metal waste from trims, buttons, zippers, and other hardware.

	5.4 Textile waste, including scraps from garment production and defective products. 5.5 Plastic waste, such as packaging materials, labels, and plastic buttons. 5.6 Solid waste from chemical treatments or dyeing processes. 5.7 Packaging waste from raw materials and finished products.
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in LET (Ready-Made Garment) manufacturing. 1.2 Sources of environmental impacts. 1.3 Ways of mitigating environmental impacts 1.4 Recyclable and non-recyclable items 1.5 Different types of waste in LET manufacturing 1.6 Hazardous waste in LET production 1.7 Green habits to reduce waste
2. Underpinning Skills	2.1 Explaining principles of green practices in LET manufacturing 2.2 Identifying sources of environmental impacts 2.3 Explaining re-cyclable and non-recyclable items 2.4 Minimizing resources use in the workplace 2.5 Identifying different types of waste in LET manufacturing 2.6 Disposing hazardous waste in LET production 2.7 Practicing green habits to reduce waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate:
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	1.1 interpreted green concepts 1.2 minimized resources use in the workplace 1.3 implemented waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: UNDERSTAND QUALITY CONTROL AND ASSURANCE	Nominal Duration: 20 Hrs.	Unit Code: SICIP-LET-QCA-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to understand quality control and assurance. It specifically includes the task of interpreting quality, interpreting quality control and quality assurance, identifying QC and QA tools and identifying roles and responsibilities of quality personnel.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret quality	1.1 Quality is defined according to industry standards, organizational and buyers' requirements. 1.2 <u>Basic quality tools and equipment</u> used for quality control and assurance is identified and described. 1.3 <u>Quality dimensions</u> are interpreted and explained. 1.4 The relationship between quality dimensions and the overall quality is understood. 1.5 Quality benchmarks and measurement criteria are identified with organizational goals for quality.
2. Interpret quality control and quality assurance	2.1 Importance of quality control systems is defined and explained. 2.2 <u>Methods of quality control</u> are identified and described. 2.3 <u>Quality control procedures</u> are identified and explained. 2.4 <u>Quality assurance parameters</u> are identified and described according to the specific needs of the product. 2.5 The difference between quality control and quality assurance is explained.
3. Identify QC and QA tools	3.1 <u>Common QC tools</u> are identified and recognized. 3.2 <u>Common QA tools</u> are identified and recognized. 3.3 Purpose and function of each QC and QA tool are explained. 3.4 Proper use of QC and QA tools is demonstrated.
4. Identify roles and responsibilities of quality personnel	4.1 The structure of the quality department is identified and described. 4.2 Reporting lines and communication flow within the quality department are identified. 4.3 <u>Core responsibilities</u> of quality personnel within the quality department are described.

Range of Variables

Variable	Range (Includes but not limited to):
1. Basic quality tools and equipment	1.1 Measuring tape 1.2 Stainless scale 1.3 3m scotch tape 1.4 White fabric duster 1.5 Needle detector machine 1.6 Moisture tester 1.7 Pilling tester 1.8 Crock meter 1.9 Button pull test machine
2. Quality dimensions	2.1 Durability 2.2 Functionality 2.3 Aesthetic appeal 2.4 Performance 2.5 Reliability 2.6 Serviceability 2.7 Conformance
3. Methods of quality control	3.1 Testing 3.2 Inspection 3.1 Monitoring and auditing
4. Quality control procedures	4.1 Leather and raw material inspection 4.2 Cutting quality control 4.3 In-process quality control (IPQC) 4.4 End-line quality inspection 4.5 Final quality control (FQC)
5. Quality assurance parameters	5.1 Dimension 5.2 Strength 5.3 Color fastness 5.4 Products safety 5.5 Embellishment 5.6 Shrinkage and spirality 5.7 Customer satisfaction
6. Common QC tools	6.1 Check Sheet 6.2 Fishbone Diagram (Cause & Effect) 6.3 Histogram 6.4 Pareto Chart 6.5 Scatter Diagram 6.6 Control Chart 6.7 Flowchart
7. Common QA tools	7.1 Standard Operating Procedures (SOPs) 7.2 Quality Manual 7.3 Lab testing 7.4 Work Instructions 7.5 Inspection Checklists

	7.6 Sampling Plans 7.7 Process Flow Diagrams 7.8 Quality Audit Reports
8. Core responsibilities	8.1 Quality Inspections and Audits 8.2 Quality Standard Compliance 8.3 Defect identification 8.4 Process Improvement 8.5 Data Analysis 8.6 Documentation and Reporting

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify the concept of quality according to industry standards, organizational, and buyer requirements. 1.2 Understand basic quality tools and equipment used for quality control and assurance. 1.3 Interpret key quality dimensions and their relationship to overall product quality. 1.4 Identify quality benchmarks and measurement criteria aligned with organizational goals. 1.5 Interpret differences between quality control (QC) and quality assurance (QA) systems. 1.6 Identify common QC and QA tools and understand their purpose and proper use. 1.7 Understand the structure, reporting lines, and communication flow of the quality department. 1.8 Identify the core roles and responsibilities of quality personnel in maintaining standards.
2. Underpinning Skills	2.1. Applying analytical skills to interpret quality dimensions and benchmarks. 2.2. Performing identification and explanation of quality control and assurance methods. 2.3. Applying problem-solving skills to maintain and improve product quality. 2.4. Following established procedures for quality control and assurance processes. 2.5. Applying observation and attention to detail when inspecting products or processes. 2.6. Performing accurate documentation of quality activities and compliance with standards.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace

	3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted quality 1.2 interpreted quality control and quality assurance 1.3 identified QC and QA tools 1.4 identified roles and responsibilities of quality personnel
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by nationally certified assessor or occupation-specific industry expert.

Unit of Competency: IDENTIFY MATERIALS, MACHINES, AND ACCESSORIES FOR LEATHER AND FOOTWEAR PRODUCTION	Nominal Duration: 25 Hrs.	Unit Code: SICIP-LET-QCA-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to identify materials machines and accessories for leather and footwear production. It specifically includes the task of interpreting leather, leather goods and footwear, identifying raw materials and chemicals, identifying cutting machine, identifying sewing machine, identifying lasting machine and identifying types of trim and accessories.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret leather, leather goods and footwear	1.1 <u>Types of leather</u> are identified. 1.2 Differences between genuine and synthetic leather are distinguished. 1.3 <u>Types of leather goods</u> are identified. 1.4 <u>Types of footwear</u> are identified.
2. Identify raw materials and chemicals	2.1 <u>Raw materials</u> used in leather and footwear production are identified and classified. 2.2 <u>Chemicals</u> used in tanning, finishing, and other processes are recognized. 2.3 Safety precautions for handling raw materials and chemicals are followed.
3. Identify cutting machine	3.1 <u>Types of cutting machines</u> are identified and classified according to their functions. 3.2 Main components of cutting machines are recognized and described. 3.3 Machine features and specifications are explained clearly. 3.4 Safety precautions for operating cutting machines are identified.
4. Identify sewing machine	4.1 <u>Types of sewing machines</u> are identified and classified according to their functions. 4.2 Main components of sewing machines are recognized and described. 4.3 Machine features and specifications are explained clearly. 4.4 Safety measures for operating sewing machines are identified.
5. Identify lasting	5.1 <u>Types of lasting machines</u> are identified and classified according to their functions.

machine	5.2 Main components of lasting machines are recognized and described. 5.3 Machine features and specifications are explained clearly. 5.4 Safety precautions for using lasting machines are identified.
6. Identify types of trim and accessories	6.1 <u>Types of trims and accessories</u> are identified and differentiated. 6.2 Uses of trims and accessories are recognized according to specification sheets. 6.3 Materials and components are classified based on quality and function. 6.4 Relevant standards for trims and accessories are interpreted.

Range of Variables

Variable	Range (Includes but not limited to):
1. Types of leather	1.1 Based on origin 1.1.1 Cowhide 1.1.2 Buffalo 1.1.3 Goat 1.1.4 Sheep 1.2 Based on tanning process 1.2.1 Chrome tanned 1.2.2 Vegetable tanned 1.2.3 Aniline tanned 1.3 Top/full grain 1.4 Split 1.5 Suede and nubuck 1.6 Bonded 1.7 Finished 1.8 Synthetic/artificial
2. Types of leather goods	2.1 Bags 1.1.1 Ladies bag 1.1.2 Gents bag 1.1.3 Unisex bag 2.2 Wallets 2.3 Belts 2.4 Jackets 2.5 Gloves 2.6 Furniture 2.7 Accessories 2.8 Luggage
3. Types of footwear	3.1 Formal Shoes 3.2 Casual Shoes 3.3 Sports Shoes

	3.4 Sandals 3.5 Slippers 3.6 Boots 3.7 Loafers 3.8 Sneakers 3.9 Moccasins 3.10 Safety Shoes
4. Raw materials	4.1 Leather (Cow, Buffalo, Goat, Sheep) 4.2 Synthetic Leather/PU Leather 4.3 Fabrics 4.4 Structural components 4.5 Rubber 4.6 TPR 4.7 EVA 4.8 PVC 4.9 Threads (nylon, polyester, cotton) 4.10 Adhesives/Glues 4.11 Fasteners 4.12 Dyes, Pigments 4.13 Oils 4.14 Fats 4.15 Foam 4.16 Padding Materials 4.17 Buckles 4.18 Eyelets 4.19 Zippers 4.20 Laces 4.21 Rivets
5. Chemicals	5.1 Primer 5.1.1 Eva primer 5.1.2 PU primer 5.1.3 PVC primer 5.2 Glue 5.3 Adhesive 5.3.1 PU adhesive 5.3.2 Water based adhesive 5.3.3 Rubber based adhesive 5.3.4 Solvent based adhesive 5.4 Finishing chemicals 5.4.1 Creams and polishes 5.4.2 Waxes 5.4.3 Dyes and pigments
6. Types of cutting machines	6.1 Die Cutting Machine 6.2 Clicker Press 6.3 Band Knife Cutting Machine 6.4 Laser Cutting Machine 6.5 Press Cutting Machine

	6.6 Hydraulic Cutting Machine
7. Types of sewing machines	7.1 Flat Bed Sewing Machine 7.2 Cylinder Bed Sewing Machine 7.3 Post Bed Sewing Machine 7.4 Walking Foot Sewing Machine 7.5 Overlock/Serger Machine 7.6 Zigzag Sewing Machine 7.7 Button Sewing Machine 7.8 Chain Stitch Machine 7.9 Industrial Lockstitch Machine 7.10 Strobel sewing machine
8. Types of lasting machines	8.1 Toe Lasting Machine 8.2 Seat and side Lasting Machine 8.3 Combination Lasting Machine 8.4 Automatic Lasting Machine 8.5 Manual Lasting Machine 8.6 Semi-Automatic Lasting Machine 8.7 Round Bottom Lasting Machine 8.8 Skiving and Lasting Combined Machine 8.9 Heat setting machine 8.10 Iron fox/sole pressing machine 8.11 Chiller and cooling machine
9. Types of trims and accessories	9.1 Eyelets 9.2 Buckles 9.3 Zippers 9.4 Velcro 9.5 Laces 9.6 Rivets 9.7 Hooks 9.8 Loops 9.9 Pads 9.10 Shank 9.11 Heel Caps 9.12 Decorative Studs 9.13 Labels 9.14 Elastic Bands

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify types of leather, leather goods, and footwear used in the industry. 1.2 Understand the differences between genuine and synthetic leather. 1.3 Identify and classify raw materials and chemicals used in leather and footwear production. 1.4 Interpret safety precautions for handling raw materials and chemicals. 1.5 Identify different types of cutting machines, their
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	components, features, and functions. 1.6 Understand safety measures required for operating cutting machines. 1.7 Identify types, components, and specifications of sewing and lasting machines. 1.8 Interpret safety precautions for operating sewing and lasting machines properly. 1.9 Identify types of trims and accessories and understand their uses, materials, and quality standards.
2. Underpinning Skills	2.1. Applying observation skills to distinguish between genuine and synthetic leather. 2.2. Performing identification and classification of raw materials and chemicals used in production. 2.3. Following safety procedures when handling chemicals, tools, and machinery. 2.4. Applying technical knowledge to interpret machine features, specifications, and safe operation practices. 2.5. Performing accurate identification of trims and accessories according to quality and function. 2.6. Demonstrating attention to detail when interpreting material and accessory standards. 2.7. Applying industry-specific terminology.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted leather, leather goods and footwear 1.2 identified raw materials and chemicals 1.3 identified cutting machine 1.4 identified sewing machine 1.5 identified lasting machine 1.6 identified types of trim and accessories
2. Methods of Assessment	Competency should be assessed by:

	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INTERPRET LEATHER, LEATHER GOODS AND FOOTWEAR MAKING PROCESS	Nominal Duration: 40 Hrs.	Unit Code: SICIP-LET-QCA-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret leather, leather goods and footwear making process. It specifically includes the tasks of identifying leather making process, identify leather goods making process and identifying footwear making process.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify leather making process	1.1 <u>Stages of the leather-making process</u> are identified and described. 1.2 <u>Tanning and finishing methods</u> are explained. 1.3 Quality checks at different stages are identified. 1.4 Safety precautions during leather making process are identified.
2. Identify leather goods making process	2.1 <u>Stages of the leather goods making process</u> are identified and described. 2.2 <u>Techniques for cutting, stitching, and finishing</u> are explained. 2.3 Quality checks at different stages of production are identified. 2.4 Safety precautions during leather goods production are identified.
3. Identify footwear making process	3.1 <u>Stages of the footwear making process</u> are identified and described. 3.2 <u>Techniques for cutting, stitching, lasting, and finishing</u> are explained. 3.3 Quality checks at different stages of footwear production are identified. 3.4 Safety precautions during footwear production are identified.

Range of Variables

Variable	Range (Includes but not limited to):
1. Stages of the leather-making process	1.1 Raw Hide Preparation 1.2 Beamhouse Operations 1.3 Tanning 1.4 Post-Tanning Operations 1.5 Dyeing

	1.6 Fat liquoring 1.7 Drying 1.8 Finishing
2. Tanning and finishing methods	2.1 Vegetable Tanning 2.2 Chrome Tanning 2.3 Aldehyde Tanning 2.4 Synthetic Tanning 2.5 Aniline Finishing 2.6 Pigmented Finishing 2.7 Suede Finishing 2.8 Embossed Finishing
3. Stages of the leather goods making process	3.1 Designing 3.2 Cutting 3.3 Splitting 3.4 Skiving 3.5 Edge coloring 3.6 Embellishment 3.7 Setting 3.8 Stitching 3.9 Assembling 3.10 Finishing 3.11 Quality Inspection
4. Techniques for cutting, stitching, and finishing	4.1 Hand Cutting 4.2 Machine Cutting 4.3 Die Cutting 4.4 Laser cutting 4.5 Hand Stitching 4.6 Machine Stitching 4.7 Overlock Stitching 4.8 Pressing 4.9 Edge Finishing 4.10 Polishing 4.11 Buffing 4.12 Gluing
5. Stages of the footwear making process	5.1 Designing 5.2 Cutting 5.3 Skiving 5.4 Stitching 5.5 Edge coloring 5.6 Embellishment 5.7 Setting 5.8 Lasting 5.9 Soling 5.10 Pressing 5.11 Polishing 5.12 Finishing 5.13 Quality Inspection
6. Techniques for cutting,	6.1 Hand Cutting

stitching, lasting, and finishing	6.2 Machine Cutting 6.3 Die Cutting 6.4 Laser cutting 6.5 Hand Stitching 6.6 Machine Stitching 6.7 Strobel Stitching 6.8 Lasting by Hand 6.9 Lasting by Machine 6.10 Buffing 6.11 Gluing 6.12 Out sole attaching 6.13 Pressing 6.14 Chiller passing 6.15 Edge Finishing 6.16 Polishing 6.17 Packing
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify stages of the leather-making process and understand key tanning and finishing methods. 1.2 Interpret quality check points and safety precautions during leather production. 1.3 Identify and describe stages of the leather goods making process. 1.4 Understand techniques for cutting, stitching, assembling, and finishing leather goods. 1.5 Identify quality control steps and safety practices in leather goods production. 1.6 Identify stages of the footwear making process and describe each stage clearly. 1.7 Understand techniques used in cutting, stitching, lasting, and finishing footwear. 1.8 Identify quality checks at different stages of footwear production. 1.9 Interpret safety measures to be followed throughout the footwear manufacturing process.
2. Underpinning Skills	2.1. Applying technical knowledge to explain tanning, cutting, stitching, lasting, and finishing techniques. 2.2. Performing identification of quality checkpoints throughout different production stages. 2.3. Applying safety practices while handling tools, machines, and materials in each process. 2.4. Following standard operating procedures to maintain consistency and product quality. 2.5. Demonstrating process awareness to coordinate workflow efficiently across production stages.

	2.6. Applying problem-solving skills to address production issues and maintain quality standards. 2.7. Performing documentation and reporting of process steps, quality results, and safety compliance.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified leather making process 1.2 identified leather goods making process 1.3 identified footwear making process
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: IDENTIFY LEATHER, LEATHER GOODS AND FOOTWEAR FAULTS AND DEFECTS	Nominal Duration: 30 Hrs.	Unit Code: SICIP-LET-QCA-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to identify leather, leather goods and footwear faults and defects. It specifically includes the task of identifying types of faults and defects, identifying leather faults and defects, identifying leather goods faults and defects, identifying footwear faults and defects and identifying trim and packing faults and defects.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify types of faults and defects	1.1 Causes of faults and defects are recognized and explained. 1.2 Defects are classified according to severity and type. 1.3 Faults and defects are classified.
2. Identify leather faults and defects	2.1 <u>Common leather faults and defects</u> are identified and recognized. 2.2 Causes of leather faults and defects are explained. 2.3 Leather faults and defects are classified. 2.4 Solutions to repair or replace leather faults are identified.
3. Identify leather goods faults and defects	3.1 <u>Common faults and defects in leather goods</u> are identified and recognized. 3.2 Causes of leather goods faults and defects are explained. 3.3 Leather goods faults and defects are classified. 3.4 Solutions to repair or replace leather goods faults are identified.
4. Identify footwear faults and defects	4.1 <u>Common faults and defects in footwear</u> are identified and recognized. 4.2 Causes of footwear faults and defects are explained. 4.3 Footwear faults and defects are classified. 4.4 Solutions to repair or replace footwear faults are identified.
5. Identify trims and packing faults and defects	5.1 <u>Common trims and packing faults</u> are identified. 5.2 Trims and packing faults and defects are classified. 5.3 Solutions to repair or replace trims and packing faults are identified.

Range of Variables

Variable	Range (Includes but not limited to):
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1. Common leather faults and defects	1.1 Loose leather 1.2 Loose grain 1.3 Growth mark 1.4 Vain mark 1.5 Pin mark 1.6 Grain damage 1.7 Wrinkles and creases 1.8 Scars and cuts 1.9 Flay cuts 1.10 Smell 1.11 Strength 1.12 Holes or insect bites 1.13 Color variations 1.14 Stains or discoloration 1.15 Stretch marks 1.16 Tick marks
2. Common faults and defects in leather goods	1.1 Grain mismatch 1.2 Color shade variation 1.3 Stitching defects 1.4 Folding defects 1.5 Loose threads 1.6 Poor edge finishing 1.7 Open leather defects 1.8 Looseness of leather 1.9 Misaligned parts 1.10 Seam defects 1.11 Scratches 1.12 Trims and accessories defects 1.13 Embossing defects 1.14 Dirty defects
2. Common faults and defects in footwear	3.1 Bonding defects 3.2 Stitching defects 3.3 Twisted shoes 3.4 Leather open defects 3.5 Misaligned sole or heel 3.6 Measurement variations 3.7 Incorrect sizing or fitting 3.8 Color mismatch 3.9 Uneven finishing or polishing 3.10 Excess glue marks 3.11 Bad smell 3.12 Rocking 3.13 Trims and accessories defects 3.14 Embossing defects 3.15 Dirty defects
3. Common trims and packing faults	3.1 Rusty or broken trims 3.2 Loose or missing trims

	3.3 Damaged trims 3.4 Poorly attached labels or logos 3.5 Discolored metal fittings 3.6 Cracked or peeling trims 3.7 Loose or misaligned elastic/bands 3.8 Faulty snaps or buttons 3.9 Packing defects 3.9.1 Wrong size or style packed 3.9.2 Incorrect labeling 3.9.3 Poor box/carton quality 3.9.4 Inadequate stuffing or wrapping 3.9.5 Damaged packaging materials 3.9.6 Improper sealing or taping 3.9.7 Missing tags or barcodes 3.9.8 Dirty or stained boxes 3.9.9 Inaccurate quantity in carton
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify different types of faults and defects and understand their causes and classifications. 1.2 Interpret severity levels and types of defects in materials and finished products. 1.3 Identify common faults and defects in leather and understand their causes and classifications. 1.4 Identify common faults and defects in leather goods and understand their causes and classifications. 1.5 Interpret appropriate corrective or replacement methods for leather goods defects. 1.6 Identify common faults and defects in footwear, along with their causes and classifications. 1.7 Identify common trims and packaging faults and determine corrective or replacement solutions.
2. Underpinning Skills	2.1 Performing identification and classification of faults and defects across leather, goods, and footwear. 2.2 Applying analytical skills to determine causes and severity of defects in materials and products. 2.3 Applying problem-solving skills to suggest appropriate repair or replacement solutions. 2.4 Following standard procedures for defect classification and quality assessment. 2.5 Applying communication skills to report and document detected faults effectively. 2.6 Performing evaluation of trims and packaging quality to ensure compliance with standards. 2.7 Demonstrating attention to detail and precision in quality control and defect analysis.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety

	3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified types of faults and defects 1.2 identified leather faults and defects 1.3 identified leather goods faults and defects 1.4 identified footwear faults and defects 1.5 identified trim and packing faults and defects
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRYOUT SECTION WISE QUALITY CONTROL	Nominal Duration: 100 Hrs.	Unit Code: SICIP-LET-QCA-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out section wise quality control. It specifically includes the task of performing quality control in sample, performing quality control in cutting, performing quality control in embellishment, performing quality control in sewing and performing quality control in lasting and finishing.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Perform quality control in sample	1.1 Approved samples are reviewed and used as quality references. 1.2 <u>Product measurements</u> are checked against the approved sample. 1.3 Design details and specifications are verified for accuracy. 1.4 Materials used in sample are inspected for quality and consistency. 1.5 <u>Finishing quality</u> is examined to ensure smoothness and appearance. 1.6 Product functionality is tested according to standards. 1.7 Findings and deviations are recorded and reported for corrective action.
2. Perform quality control in cutting	2.1 Standard operating procedures (SOP) and specifications are identified. 2.2 Materials are inspected before cutting to ensure suitability. 2.3 Cutting tools and machines are checked for proper condition and setup. 2.4 Cut pieces are measured and compared with approved patterns. 2.5 Cutting faults and defects are detected and recorded. 2.6 Findings are documented and reported for corrective action.
3. Perform quality control in embellishment	3.1 Design layouts and color patterns are checked against approved samples. 3.2 <u>Printing, embossing and embroidery materials</u> are inspected before use. 3.3 <u>Alignment, spacing, and consistency of prints or stitches</u> are verified. 3.4 Defects such as color mismatch, misprint, or broken threads are detected and recorded.

	3.5 Finished printed and embroidered products are inspected for overall appearance and quality. 3.6 Non-conformities are reported and corrective actions are taken.
4. Perform quality control in sewing	4.1 Sewing quality standards and specifications are identified and reviewed. 4.2 Needles, thread types, stitches and seams are checked for compliance with approved samples. 4.3 Sewing machines and tools are inspected for proper setup and operation. 4.4 <u>Leather goods/leather footwear components</u> are examined for defects. 4.5 Measurements and alignment of sewn pieces are verified. 4.6 Findings are reported and corrective actions are implemented.
5. Perform quality control in lasting and finishing	5.1 <u>Lasting tools</u> are inspected for proper lasting and assembly according to specifications. 5.2 Surface finishing, polishing, and trimming are checked for consistency and quality. 5.3 Measurements and alignment of lasted products are verified against approved samples. 5.4 Chemical mixing, adhesion and component attachment are examined. 5.5 Defects such as uneven finishing, poor lasting, or material damage are detected and recorded. 5.6 Findings are reported and corrective actions are implemented to maintain quality standards.

Range of Variables

Variable	Range (Includes but not limited to):
1. Product measurements	Leather Goods 1.1 Length 1.2 Width 1.3 Depth / Thickness 1.4 Strap or handle length 1.5 Pocket or compartment dimensions 1.6 Zipper length 1.7 Closure positioning 1.8 Seam allowance 1.9 Hardware placement (buckles, rivets, buttons) 1.10 Overall shape and symmetry Footwear 1.11 Length of the shoe (Toe, Vamp, Quarter) 1.12 Width (forefoot and heel) 1.13 Height of the heel, Back height 1.14 Toe spring

	1.15 Toe box and counter stiffener dimensions 1.16 Insole 1.17 Outsole 1.18 Strap or lace length 1.19 Heel counter and arch positioning
2. Finishing quality	2.1 Surface smoothness 2.2 Color uniformity 2.3 Gloss / Shine consistency 2.4 Edge finishing 2.5 Polishing quality 2.6 Seam neatness 2.7 Crease / wrinkle absence 2.8 Scratch and mark free 2.9 Trim and accessory alignment 2.10 Overall aesthetic appearance
3. Printing, embossing and embroidery materials	3.1 Heat transfer foils 3.2 Ink and pigments 3.3 Embroidery threads (cotton, polyester, metallic) 3.4 Embroidery backing / stabilizers 3.5 Embossing plates or dies 3.6 Stencils and screens for printing 3.7 Adhesives for applique or embroidery 3.8 Leather-safe paints and dyes 3.9 Heat press films / transfer papers 3.10 Metallic or decorative studs for embossing/embroidery accents
4. Alignment, spacing, and consistency of prints or stitches	4.1 Correct placement of logos, labels, or motifs 4.2 Uniformity of color in prints or embossing 4.3 Straightness of stitches or printed lines 4.4 Even spacing between stitches or design elements 4.5 Consistent stitch length or print thickness 4.6 Symmetry of patterns or logos 4.7 Alignment of seams with printed or embroidered designs 4.8 Positioning of decorative elements relative to product edges 4.9 Consistency of thread tension or ink application 4.10 Overall visual balance and aesthetic appearance
5. Leather goods/leather footwear components	Leather Goods Components 5.1 Main leather body / panels 5.2 Lining material 5.3 Zippers 5.4 Buckles 5.5 Buttons / Snaps 5.6 Rivets 5.7 Straps / Handles 5.8 Threads / Stitching 5.9 Labels / Tags / Logos

	5.10 Decorative studs or appliques Leather Footwear Components 5.11 Upper (vamp, quarters, tongue) 5.12 Lining (Interlining, foam and reinforcement) 5.13 Insole 5.14 Midsole 5.15 Outsole 5.16 Heel 5.17 Eyelets / Hooks / Lace holes 5.18 Laces or straps 5.19 Stitching / Thread 5.20 Decorative elements (buckles, studs, embossing)
6. Lasting tools	6.1 Shoe lasts / molds 6.2 Lasting pinchers 6.3 Hammers / Mallets 6.4 Clamps / Vices 6.5 Stretching tools 6.6 Adhesive applicators / Brushes 6.7 Adhesive mixture 6.8 Sole pressing machines 6.9 Heel setters 6.10 Alignment gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify quality control requirements and standards for samples, cutting, embellishment, sewing, and finishing stages. 1.2 Understand the use of approved samples as reference points for quality verification. 1.3 Interpret measurement, design, and material specifications to ensure consistency with approved standards. 1.4 Identify common faults and defects during cutting, sewing, and finishing operations. 1.5 Understand the importance of inspecting materials, tools, and equipment before and during production. 1.6 Interpret methods for detecting, recording, and reporting non-conformities and deviations. 1.7 Understand inspection techniques for design alignment, stitching, finishing, and product functionality. 1.8 Apply corrective actions to maintain product quality throughout the production process. 1.9 Recognize the role of documentation and communication in quality control and assurance.
2. Underpinning Skills	2.1 Performing inspection and measurement to verify product quality against approved samples. 2.2 Applying attention to detail when checking design

	accuracy, materials, and finishing quality. 2.3 Applying analytical skills to identify deviations, defects, and non-conformities during production. 2.4 Performing documentation and reporting of findings for corrective and preventive action. 2.5 Applying problem-solving skills to recommend and implement quality improvements. 2.6 Following standard operating procedures (SOPs) and safety practices during all quality control activities. 2.7 Applying teamwork and communication skills to coordinate with production teams for quality assurance.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 performed quality control in sample 1.2 performed quality control in cutting 1.3 performed quality control in embellishment 1.4 performed quality control in sewing 1.5 performed quality control in lasting and finishing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY ROOT CAUSE ANALYSIS TECHNIQUES	Nominal Duration: 40 Hrs.	Unit Code: SICIP-LET-QCA-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply root cause analysis techniques. It specifically includes the task of analyzing data and identifying root causes, implementing root cause analysis techniques, implementing corrective and preventive actions (CAP/PAP) and prepare report analysis results.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Analyze data and identify root causes	1.1 Data related to production, quality, and defects are collected and organized. 1.2 Patterns, trends, and deviations in the data are analyzed. 1.3 Potential causes of problems and defects are identified using root cause analysis techniques. 1.4 Relationships between causes and effects are examined and documented. 1.5 Root causes are reported and used to recommend corrective actions.
2. Implement root cause analysis techniques	2.1 <u>Root cause analysis techniques</u> are identified and selected for use. 2.2 Data on defects, errors, or inefficiencies are collected and reviewed. 2.3 Root causes of identified problems are determined and documented. 2.4 Findings are validated with production records and observations. 2.5 Corrective actions are suggested based on analysis results. 2.6 Results of root cause analysis are reported to relevant personnel for implementation.
3. Implement corrective and preventive actions (CAP/PAP)	3.1 <u>Corrective and preventive actions</u> are identified based on root cause analysis findings. 3.2 Approved corrective measures are implemented to address identified issues. 3.3 Preventive actions are applied to avoid recurrence of problems. 3.4 Effectiveness of implemented actions is monitored and evaluated. 3.5 Deviations or non-conformities after implementation are detected and recorded.

	3.6 Adjustments are made as needed to improve outcomes. 3.7 Actions taken are documented and communicated to relevant personnel.
4. Prepare report analysis results	4.1 Data from root cause analysis is compiled and organized. 4.2 Key findings and contributing factors are summarized. 4.3 Corrective and preventive actions are documented. 4.4 Reports are formatted according to organizational standards. 4.5 Analysis results are interpreted and presented clearly. 4.6 Reports are reviewed and validated for accuracy. 4.7 Final reports are submitted to relevant personnel for decision-making and follow-up.

Range of Variables

Variable	Range (Includes but not limited to):
1. Root cause analysis techniques	1.1 5 Whys Analysis 1.2 Fishbone Diagram (Ishikawa) 1.3 Pareto Analysis 1.4 Failure Mode and Effects Analysis (FMEA) 1.5 Cause-and-Effect Charting 1.6 Check Sheets 1.7 Scatter Diagrams 1.8 Brainstorming Sessions
2. Corrective and preventive actions	2.1 Implement additional training for workers 2.2 Repair or rework defective products 2.3 Update standard operating procedures (SOPs) 2.4 Conduct preventive maintenance on machinery 2.5 Introduce quality checkpoints during production 2.6 Adjust machine settings or calibrations 2.7 Replace worn-out tools or components 2.8 Improve material handling and storage 2.9 Revise inspection and testing methods 2.10 Modify design or process to eliminate recurring defects
3. Reports	3.1 Root Cause Analysis Report 3.2 Quality Control Report 3.3 Defect / Rework Report 3.4 Machine Downtime Report 3.5 Production Report 3.6 Inventory / Material Usage Report 3.7 Corrective and Preventive Action (CAPA) Report 3.8 Inspection / Audit Report 3.9 Productivity Report

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify data sources related to production, quality, and defects for analysis. 1.2 Understand methods for organizing and analyzing data to identify patterns, trends, and deviations. 1.3 Interpret root cause analysis techniques and their application in identifying production problems. 1.4 Understand the relationship between causes and effects in production and quality issues. 1.5 Identify and implement appropriate corrective and preventive actions (CAP/PAP) based on analysis findings. 1.6 Interpret methods to monitor, evaluate, and adjust corrective and preventive actions for effectiveness. 1.7 Understand documentation and reporting requirements for analysis results and corrective actions. 1.8 Identify procedures for preparing, formatting, and validating analysis and CAP/PAP reports. 1.9 Interpret how analysis reports support management decisions and continuous improvement.
2. Underpinning Skills	2.1 Performing data collection, organization, and interpretation to identify trends and deviations. 2.2 Applying analytical and critical thinking skills to conduct root cause analysis effectively. 2.3 Applying problem-solving skills to develop and implement corrective and preventive actions. 2.4 Performing validation and verification of findings through observation and production records. 2.5 Applying evaluation skills to monitor the effectiveness of implemented actions. 2.6 Following organizational standards and procedures for preparing and submitting reports.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 analyzed data and identify root causes 1.2 implemented root cause analysis techniques 1.3 implemented corrective and preventive actions (CAP/PAP) 1.4 prepared report analysis results
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM FINAL INSPECTION AND AUDIT	Nominal Duration: 55 Hrs.	Unit Code: SICIP-LET-QCA-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform final inspection and audit. It specifically includes the task of preparing for final inspection, conducting final product inspection and interpreting technical audit system.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for final inspection	1.1 <u>Final inspection requirements</u> and checklists are reviewed and understood. 1.2 Products are organized and prepared for inspection. 1.3 Relevant tools, equipment, and measuring devices are checked and set up. 1.4 Approved samples or standards are collected for comparison. 1.5 Inspection environment is arranged to ensure accurate assessment. 1.6 Documentation and records required for inspection are gathered. 1.7 Preparations are verified to ensure readiness for the final inspection.
2. Conduct final product inspection	2.1 Final products are examined against approved standards and specifications. 2.2 Measurements, dimensions, and tolerances are checked for compliance. 2.3 Visual defects such as scratches, stains, or misalignment are identified. 2.4 Functionality and performance of products are tested. 2.5 Non-conforming products are detected and segregated. 2.6 Inspection results are recorded accurately for traceability. 2.7 Findings are reported to relevant personnel for corrective action or approval.
3. Interpret technical audit system	3.1 Technical audit system objectives and requirements are identified and explained. 3.2 <u>Audit procedure checklist</u> is understood and prepared. 3.3 Data and records required for the audit are collected and organized. 3.4 Audit criteria are interpreted to assess compliance with standards.

	3.5 Findings and deviations are analyzed and documented. 3.6 Recommendations for improvements are identified based on audit results. 3.7 Audit outcomes are reported to relevant personnel for decision-making.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Final inspection requirements	1.1 Check overall appearance 1.2 Verify packaging and labeling 1.3 Check dimensions against sample 1.4 Inspect stitching and seams 1.5 Examine surface for defects 1.6 Check edges and finishing 1.7 Verify trims and hardware 1.8 Inspect sole and heel attachment (footwear) 1.9 Test zippers, buckles, and closures 1.10 Acceptable Quality Level (AQL) 1.11 Record inspection results
2. Audit procedure checklist	2.1 Product samples 2.2 Production records 2.3 Materials and inventory record 2.4 Machinery maintenance logbook 2.5 Section wise quality control 2.6 Safety and labor compliance 2.7 Company quality manuals

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify the purpose and requirements of final product inspection according to standards and checklists. 1.2 Understand preparation procedures for inspection. 1.3 Interpret the use of approved samples and reference standards for quality comparison. 1.4 Identify key inspection parameters. 1.5 Understand methods for detecting non-conforming products and maintaining traceability records. 1.6 Interpret the objectives and procedures of a technical audit system. 1.7 Identify audit criteria, analyze deviations, and document findings accurately. 1.8 Interpret how audit results and recommendations contribute.
2. Underpinning Skills	2.1 Performing preparation and organization of products, tools, and documents for final inspection. 2.2 Applying attention to detail when checking measurements, tolerances, and visual quality

	standards. 2.3 Applying analytical skills to identify non-conformities and deviations during inspection. 2.4 Performing accurate documentation and reporting of inspection findings for traceability. 2.5 Applying understanding of technical audit systems, criteria, and compliance requirements. 2.6 Following standard inspection and audit procedures to maintain consistency and reliability. 2.7 Applying communication and coordination skills to report results and support decision-making.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
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End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Quality Control & Assurance
Number of Trainees:	20

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptops	01
2.	Multimedia projector & screen	01
3.	Display boards with showing components of leather goods	04
4.	Display boards with showing components of footwear	04
5.	Materials show boards	02
6.	Swatch book	02
7.	Defects library of leather	01 set
8.	Defects library of leather goods	01 set
9.	Defects library of footwear	01 set
10.	Scanner	01
11.	Printer	01
12.	Measurement tape	20
13.	Measurement scale	20
14.	White duster fabric	20
15.	Stop watch	02
16.	Leather thickness measuring gauge	02
17.	Approved samples of goods and footwear	06

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Quality Control & Assurance the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according

to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 20 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 20 trainees.